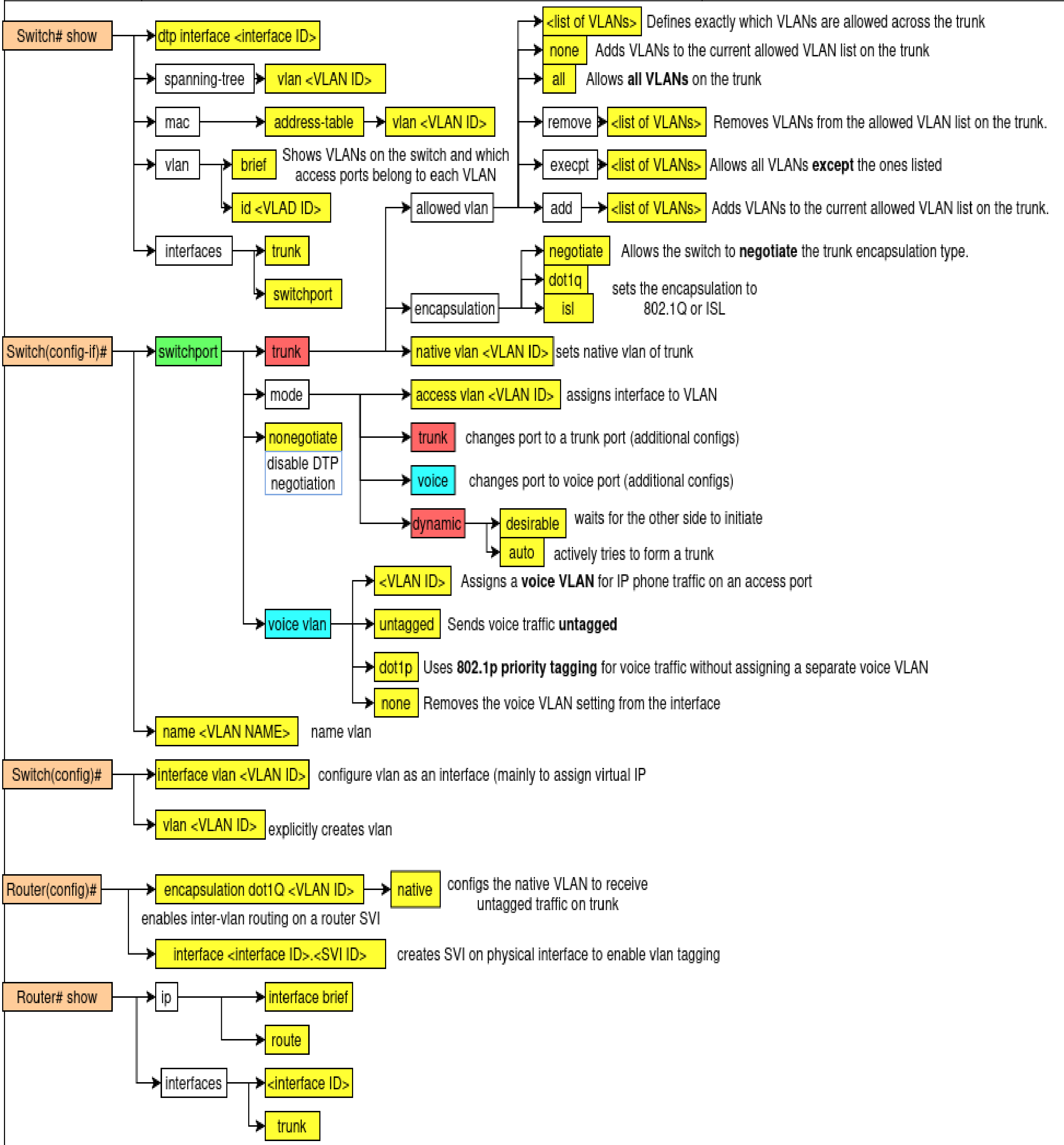
Data Link / 2 Layer

VLAN	802.1Q / Native VLAN:	Ports / DTP:
Purpose: Segments one physical LAN into multiple logical broadcast domains.	**802.1Q / Native VLAN:** Switches add an 802.1Q tag to trunk frames to identify the VLAN. Frames on the native VLAN are sent untagged. Untagged frames received on a trunk are placed in the native VLAN. Tags are removed before frames are sent to end hosts. **Inter-VLAN Routing:** Traffic cannot move between VLANs through Layer 2 switching alone. Communication between VLANs requires a Layer 3 device, such as a router or Layer 3 switch.	**Access** carries one VLAN only. **Trunk** carries multiple VLANs. **DTP** negotiates whether a link becomes an access link or trunk. **Dynamic desirable** actively tries to form a trunk. **Dynamic auto** waits for the other side to initiate. Two **dynamic auto** ports do not form a trunk.



Rapid Spanning Tree Protocol (RSTP)

Purpose: Prevents Layer 2 loops and broadcast storms by placing redundant links into a non-forwarding state when needed.

Port States:
Discarding: does not forward frames and does not learn MAC addresses; used to prevent loops while waiting or blocking.
Learning: does not forward frames yet, but starts learning source MAC addresses to build the MAC table.
Forwarding: fully operational state; forwards frames and learns MAC addresses.

Port Roles:
Root Port: the best port on a non-root switch that points toward the root bridge.
Designated Port: the port allowed to forward traffic for a segment because it has the best path from that segment to the root.
Alternate Port: a backup path toward the root bridge; normally kept in a non-forwarding state unless needed.
Disabled: administratively down or unusable, so it does not participate in STP operations.

Root Election / Path Selection Process: 1) **Elect Root Bridge** by choosing the switch with the **lowest Bridge ID**.

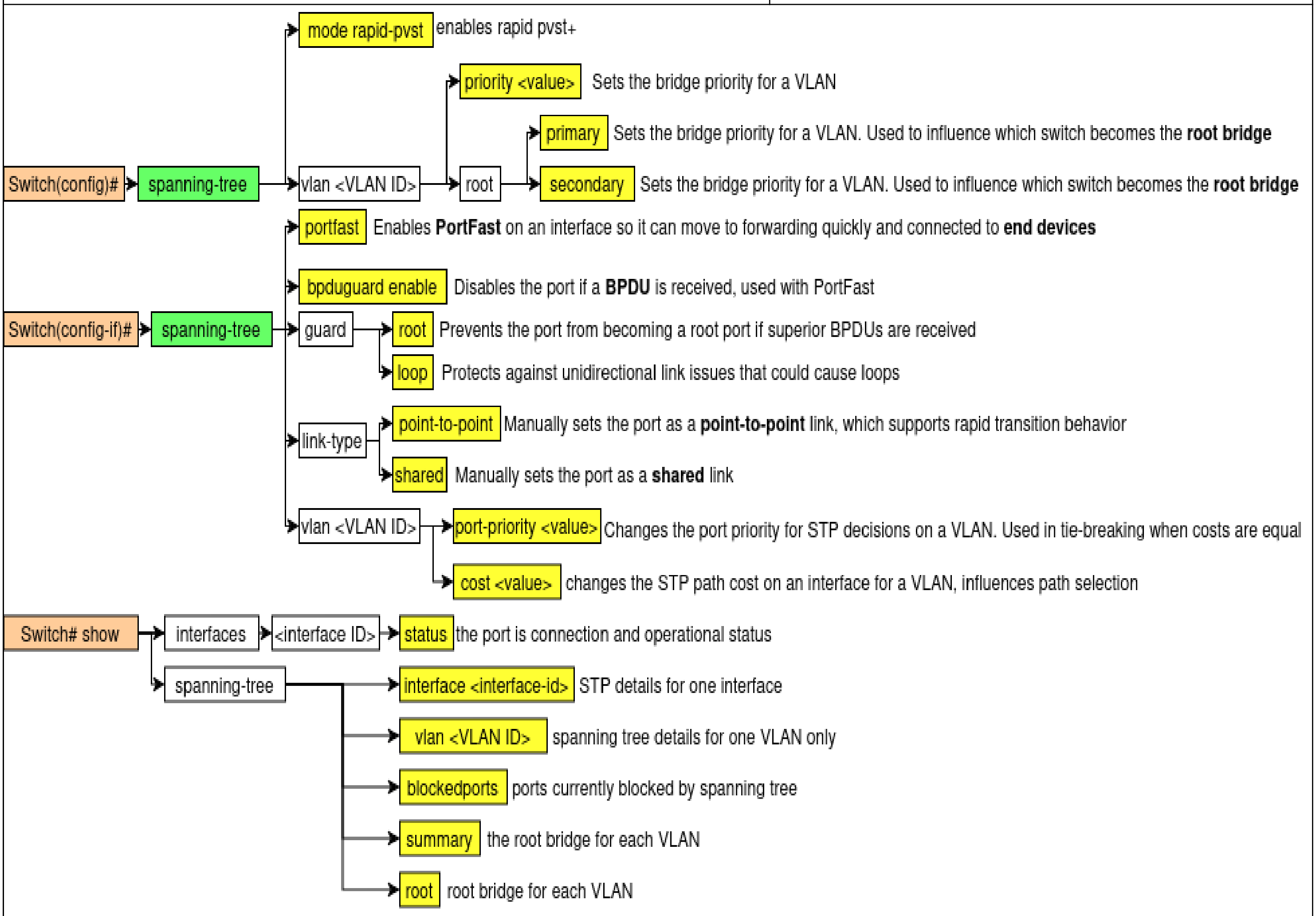
2) **Bridge ID** is compared by **priority first**, then **MAC address** if priorities are equal.

3) Each non-root switch selects one **Root Port**, which is the port with the **lowest total path cost** to the root bridge.

4) On each network segment, one **Designated Port** is chosen as the port that sends the **best BPDU** onto that segment.

5) Any other redundant ports on that segment become **Alternate** ports to prevent loops.

6) If the active path fails, an alternate port can quickly transition to forwarding.



First-Hop Redundency Protocol (FHRP)/ Host Standby Router Protocol (HSRP)

Purpose: Provides **default gateway redundancy** so hosts can still reach other networks if one router fails.

Election Behavior:
Highest **priority** becomes active. If tied, highest **IP address** wins. Next best router becomes standby. **Preemption** lets a higher-priority router retake the active role.

Failure Process:
If the standby router stops receiving **hello messages**, it assumes the active router has failed and becomes the new **active router**. It then sends a **gratuitous ARP** so hosts update the gateway MAC entry and continue forwarding with minimal disruption.

Operation Process:

1) Routers join the same **HSRP group**.

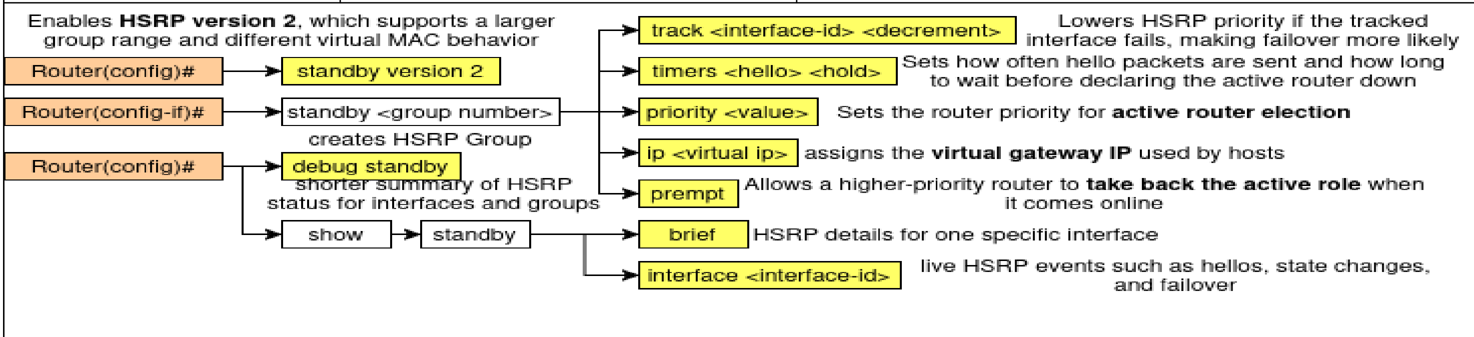
2) The group uses a **virtual IP** and **virtual MAC**.

2) One router becomes **active** and forwards traffic.

3) One router becomes **standby** and waits to take over.

4) Hosts use the **virtual IP** as their default gateway.

5) The active router sends **hello messages**.



EtherChannel / Port-Channel

Purpose: Combines multiple physical links into one **logical link** to increase bandwidth and provide redundancy.

Channel Types:
1) Static configuration
2) Port Aggregation Protocol (PagP)
3) Link Aggregation Control Protocol (LACP) – IEEE 802.3ad

Negotiation Modes:
PAgP desirable: actively tries to form the channel.
PAgP auto: waits for the other side to initiate.
LACP active: actively tries to form the channel.
LACP passive: waits for the other side to initiate.

Failure / STP:
If one link fails, traffic uses the remaining links. The EtherChannel fails only if all member links fail. STP treats it as one logical link, reducing blocked redundant links.

Load Balancing / Operation:
Traffic is balanced across member links based on a hash such as source/destination MAC or IP. Frames in the same flow usually stay on the same physical link to avoid out-of-order delivery.

Negotiation Modes:

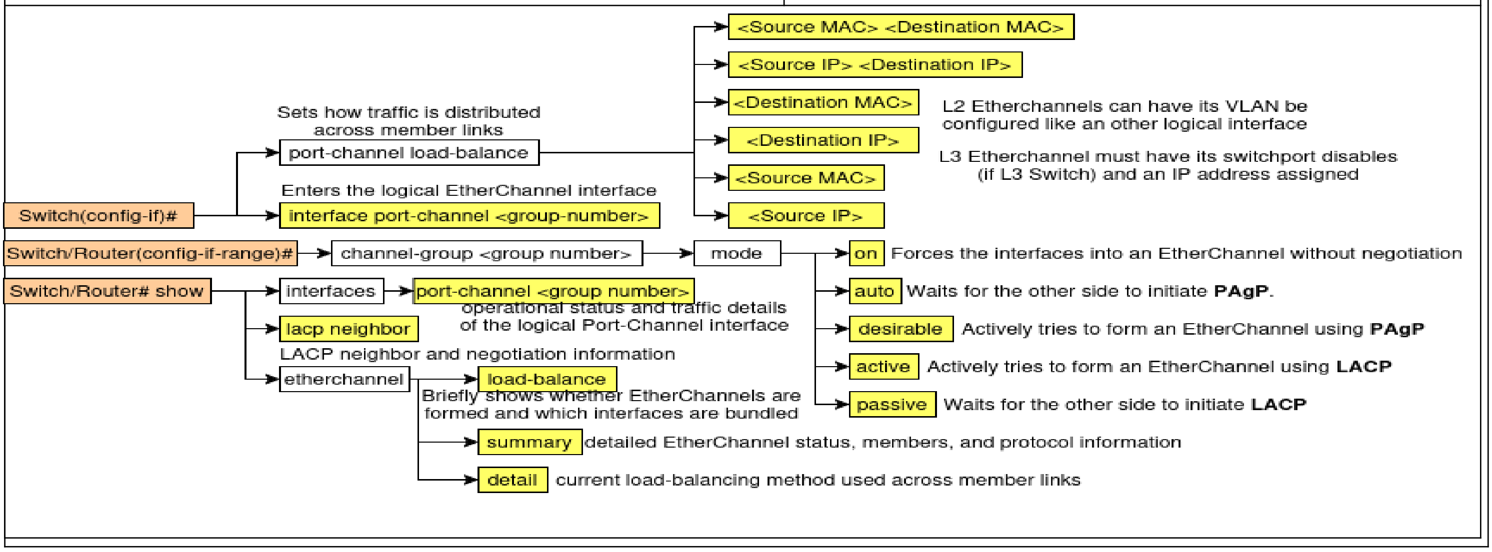
PAgP: desirable initiates, **auto** waits.

LACP: active initiates, **passive** waits.

Formation Behavior:

Forms with **on + on**, **desirable + auto**, and **active + passive**.

Does not form with **auto + auto** or **passive + passive**.



Address Resolution Protocol (ARP)	Packet Types:
Purpose: Resolves an IPv4 address to a MAC address on the local network	**Request** is a Layer 2 broadcast asking who owns the target IP. **Reply** is usually a unicast carrying the sender's MAC address. ARP is carried directly in an Ethernet frame, not inside an IP packet.

Network / 3 Layer

General Routing	Longest Prefix Match:	Route Aggregation / Summarization:	Wildcard Mask:
	If multiple routes match a destination, the router chooses the route with the **most matching network bits**.	Multiple contiguous routes can be combined into one **summary route** using the common leftmost bits, reducing routing table size.	An inverted subnet mask used to match address ranges. **0** means bits must match, **1** means bits can vary.

Open Shortest Path First (OSPF)
Purpose: Link-state routing protocol that learns topology and uses **Dijkstra SPF** to choose lowest-cost paths

OSPF can advertise a **default route**. **Passive interfaces** stop hello packets and neighbor formation while still advertising the connected network.

Cost:
Path cost is based on interface bandwidth. Default formula: **cost = reference bandwidth / interface bandwidth**. Cost can be changed by adjusting **reference bandwidth**, **interface bandwidth**, or manual **ip ospf cost**. Interfaces at **100 Mbps or higher** may all appear as cost 1 unless the reference bandwidth is increased.

Packet Types in the Process

Hello: discovers neighbors and maintains adjacency.
DBD: summarizes the link-state database.
LSR: asks for missing or newer LSAs.
LSU: sends the requested or updated LSAs.
LSAck: acknowledges receipt of LSAs.

Areas / Metrics:
OSPF uses **areas** to reduce database size and processing. Default **administrative distance** is **110**. Supports **equal-cost load balancing**; default maximum paths is commonly **4**.

Router ID / Roles:
Router ID is the highest **loopback IP**, otherwise highest active interface IP. On multiaccess links, routers elect a **DR** and **BDR** by **priority**, then Router ID.

OSPF Process / Phases:

1) Discover neighbors - Routers start by sending **Hello packets** to the OSPF multicast address to find other OSPF routers on the same link.

2) Down state - No hellos have been received from a neighbor yet, so the router does not know of any active OSPF neighbor on that interface.

3) Init state - A hello packet is received from a neighbor, but the receiving router does not yet see its own Router ID listed in that neighbor's hello packet.

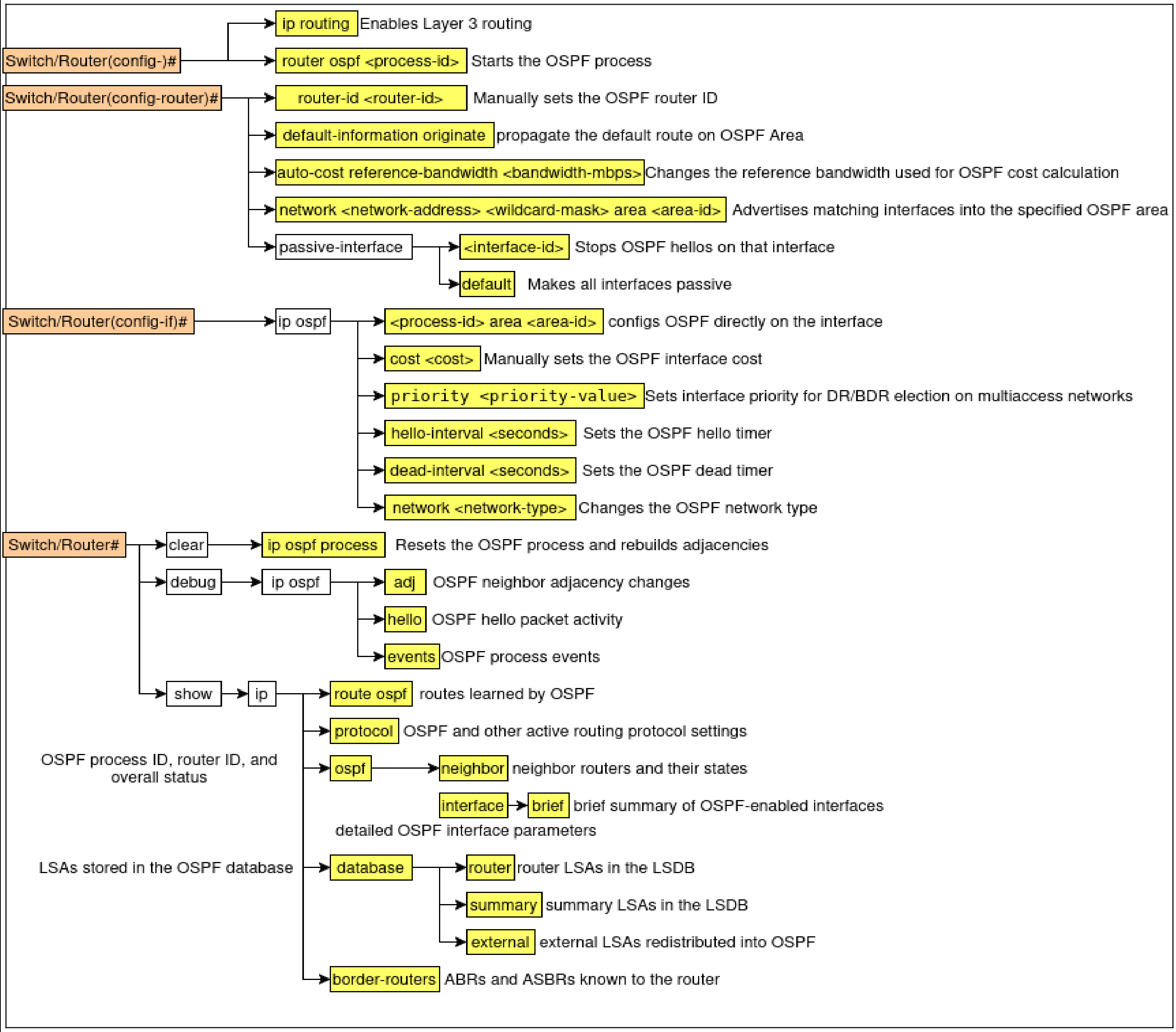
4) 2-Way state - The router now sees its own Router ID in the neighbor's hello packet, confirming **bidirectional communication**. On multiaccess networks, this is also where **DR and BDR election** takes place. Routers that are neither DR nor BDR become **DROTHERs**.

5) ExStart state - Routers decide which one will start the database exchange and establish the master-slave relationship for synchronization.

6) Exchange state - Routers exchange **DBD (Database Description) packets**, which are summaries of their link-state databases, so each router can see what information the other router has.

7) Loading state - If a router finds missing or outdated LSAs after reading the DBDs, it sends an **LSR (Link-State Request)**.
The neighbor responds with an **LSU (Link-State Update)**, and the receiving router confirms with an **LSAck (Link-State Acknowledgment)**.

8) Full state - The routers have fully synchronized their **link-state databases** and are fully adjacent. At this point, they have enough topology information to run the **SPF algorithm**. **9) Calculate best paths -** Each router runs **Dijkstra's Shortest Path First algorithm** on the synchronized database to calculate the lowest-cost routes to remote networks.



Internet Control Message Protocol (ICMP)

Purpose: Used by IP devices to send error reports and operational / diagnostic messages.

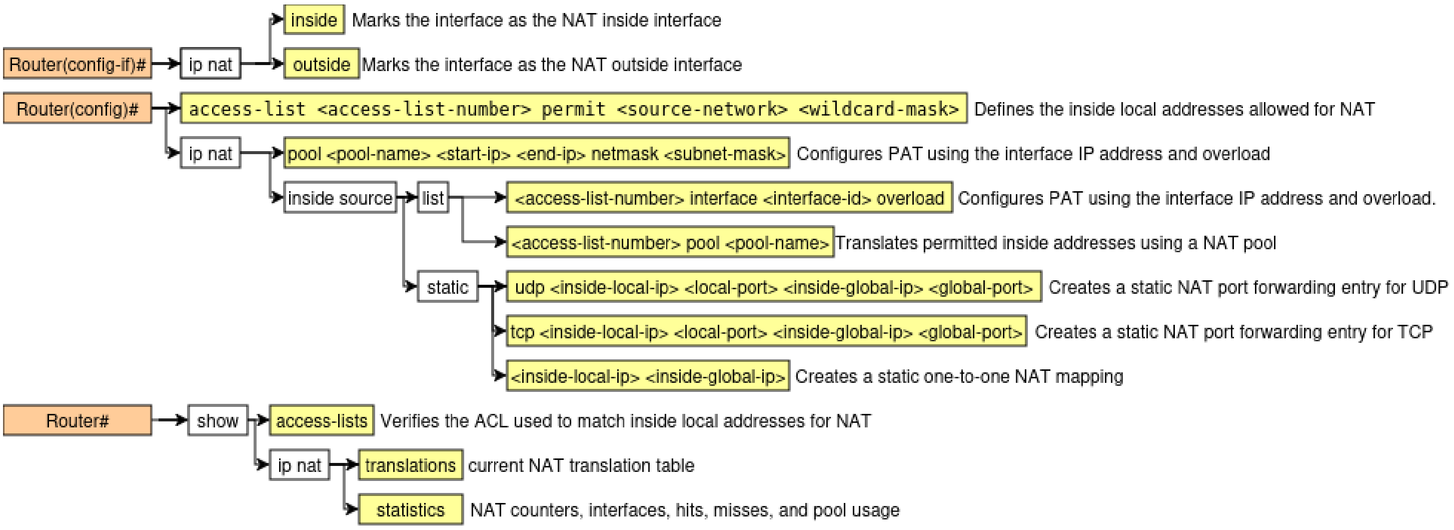
Common Types:
Type 3 Destination Unreachable
Type 11 Time Exceeded
Type 8 / 0 Echo Request / Echo Reply
Type 13 / 14 Timestamp Request / Reply

Message Categories:
Error messages: report delivery problems.
Query / Reply messages: used for testing and diagnostics.

ICMP Type 3 Destination Unreachable Codes

0 Destination network unreachable
1 Destination host unreachable
2 Destination protocol unreachable
3 Destination port unreachable
4 Fragmentation required, and DF flag set
5 Source route failed
6 Destination network unknown
7 Destination host unknown
8 Source host isolated
9 Network administratively prohibited
10 Host administratively prohibited
11 Network unreachable for ToS
12 Host unreachable for ToS
13 Communication administratively prohibited
14 Host precedence violation
15 Precedence cutoff in effect

<u>**Network Address Translation (NAT)**</u> **Purpose:** Translates private IP addresses to public IP addresses for external communication.	**Private addresses**: 10.0.0.0 – 10.255.255.255 172.16.0.0.0 – 172.31.255.255 192.168.0.0 – 192.168.255.255	**NAT Types:** **Static NAT:** fixed 1-to-1 private to public mapping. **Dynamic NAT:** maps private addresses to a public pool. **PAT / NAT Overload:** maps many private hosts to one public IP using different port numbers.



Transport / 4 Layer

TCP - Provides reliable data delivery **Purpose:** Reliable, connection-oriented transport protocol.	**Services:** Process addressing with **port numbers**, **connection establishment/termination**, **flow control**, **error recovery**, and **congestion control**.	**Connection Closing:** 1) One side sends **FIN**. 2) The other side replies **ACK** and sends its own **FIN**. 3) The first side replies **ACK**.	**Connection Establishment:** 1) Client sends **SYN** with initial sequence number. 2) Server replies **SYN-ACK** with its own initial sequence number. 3) Client sends **ACK** to complete the three-way handshake.

TCP Sliding Window 1) The **sliding window** is the number of **bytes** the sender can transmit before needing an **ACK**. 2) TCP uses it for **flow control**, so the sender does not send more data than the receiver can handle. 3) The receiver advertises a **window size**, which tells the sender how much buffer space is available. 4) The sender can send all bytes that fit within the current window without stopping after every segment. 5) As data is sent, the available space in the sender's current window decreases. 6) When an **ACK** is received, the sender knows some bytes were successfully received. 7) The ACK number tells the sender the **next byte expected**, meaning all earlier bytes were received.	TCP Error Control 1) TCP uses **sequence numbers** to track the order of bytes sent. 2) The receiver uses **ACKs** to confirm which bytes were received successfully. 3) If an ACK is not received within the **Retransmission Timeout (RTO)**, TCP assumes the segment or ACK was lost. 4) The sender then **retransmits** the missing segment. 5) If segments arrive **out of order**, TCP uses sequence numbers to place them in the correct order. 6) If a segment is **duplicated**, TCP

8) After the ACK arrives, the sender's allowed sending range moves forward, so the window **slides**. 9) If the receiver advertises a **larger window**, the sender can send more data. 10) If the receiver advertises a **smaller window**, the sender must slow down. 11) If the receiver advertises a **window size of 0**, the sender must stop sending until the window opens again. 12) Both sender and receiver maintain their own window and buffer information during communication. 13) This improves efficiency because TCP can send **multiple segments at once** instead of waiting for an ACK after every segment. 14) Example: if the window is **3000 bytes**, the sender may send bytes **1–3000**. If it later receives **ACK 2001**, the window slides forward and the sender can continue from byte **2001** onward. In short, the sliding window is a **moving byte range** that controls how much unacknowledged data can be in transit at one time.	can detect it because the sequence number was already received. 7) If data is corrupted, the **checksum** helps detect the error, and the bad segment is discarded. 8) Since the corrupted segment is not acknowledged properly, the sender retransmits it. 9) This process allows TCP to recover from **lost, damaged, duplicate, or out-of-order** segments. 10) Overall, TCP error control provides **reliable delivery** by detecting problems and retransmitting data when needed.

UDP **Purpose:** Fast, connectionless transport protocol.	**Characteristics:** Provides **low-overhead**, **best-effort** delivery with no connection setup, no reliability, no retransmission, and no flow control. It is **datagram-oriented**.

Application / 7 Layer

Dynamic Host Configuration Protocol (DHCP) **Purpose:** Dynamically assigns IP addressing information to hosts.	**DHCP Relay Agent:** A relay agent forwards client DHCP broadcasts to a DHCP server on another network by replacing the destination address with the server's address.	**Ports / Transport:** Uses **UDP**. **Server:** port **67** **Client:** port **68**

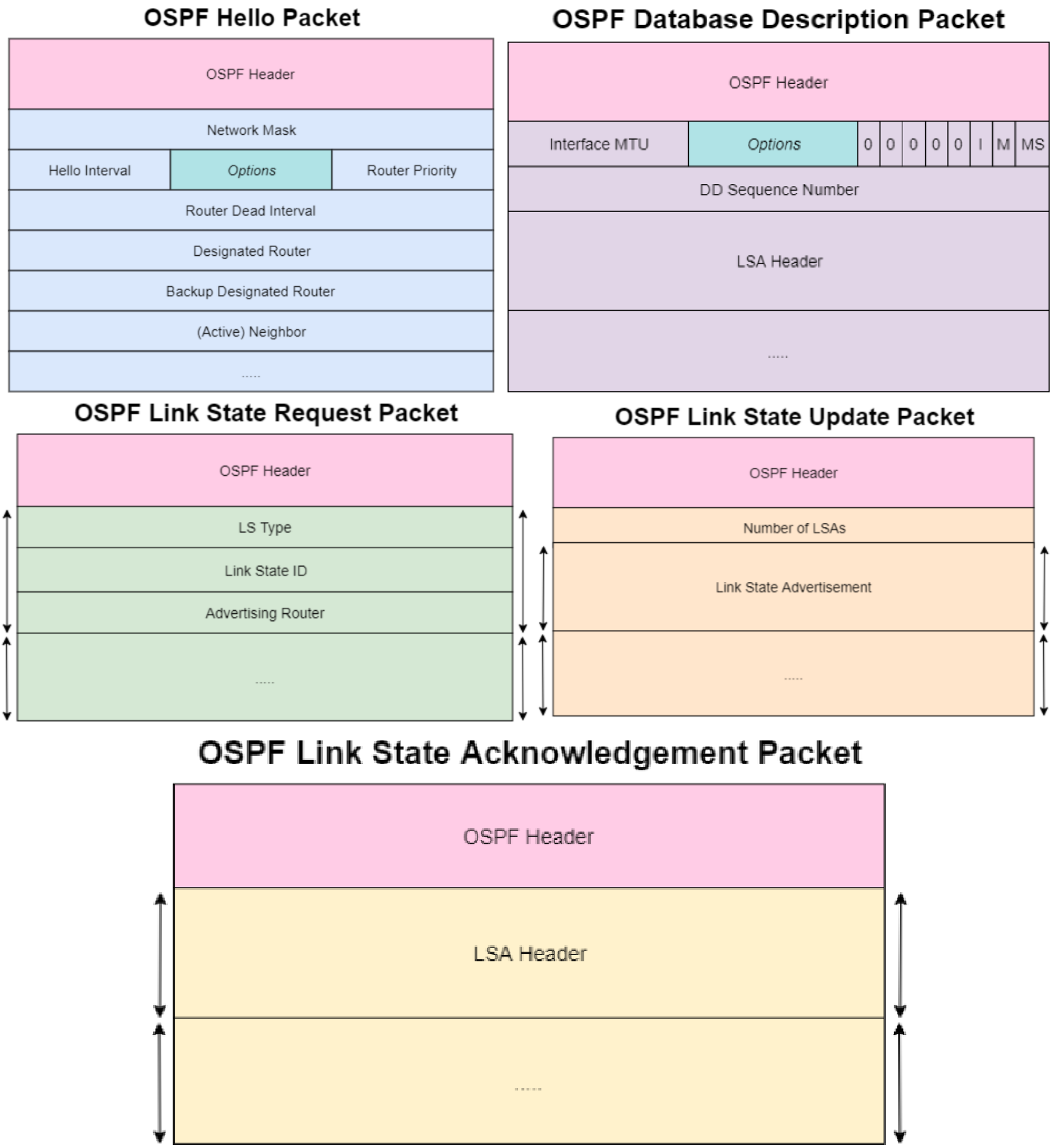
TCP Error Control

1. TCP uses **sequence numbers** to track the order of bytes sent.
2. The receiver uses **ACKs** to confirm which bytes were received successfully.
3. If an ACK is not received within the **Retransmission Timeout (RTO)**, TCP assumes the segment or ACK was lost.
4. The sender then **retransmits** the missing segment.
5. If segments arrive **out of order**, TCP uses sequence numbers to place them in the correct order.
6. If a segment is **duplicated**, TCP can detect it because the sequence number was already received.
7. If data is corrupted, the **checksum** helps detect the error, and the bad segment is discarded.
8. Since the corrupted segment is not acknowledged properly, the sender retransmits it.
9. This process allows TCP to recover from **lost, damaged, duplicate, or out-of-order** segments.
10. Overall, TCP error control provides **reliable delivery** by detecting problems and retransmitting data when needed.

DNS **Purpose:** Resolves domain names to IP addresses so hosts can find network resources	**Common Records:** **A:** maps name to IPv4 address. **AAAA:** maps name to IPv6 address. **CNAME:** alias to another name. **MX:** mail server for a domain.	**Resolution Process:** 1. Client sends a DNS query for a domain name. 2. DNS server replies with the matching

by name.	**NS:** authoritative name server. **PTR:** reverse lookup from IP to name. **TXT:** text information.	record if known. 3. If not known, the server may query other DNS servers to find the answer. 4. The final reply returns the requested record, such as an IP address.
Ports / Transport: Uses **UDP 53** for most queries and replies. Uses **TCP 53** for zone transfers and sometimes large responses.		

OSPF Packet Types

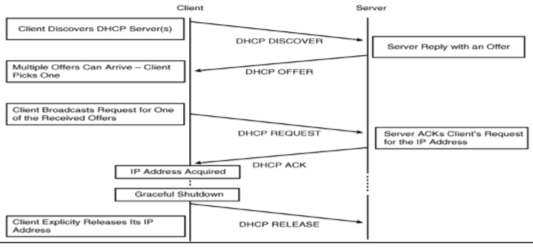


Application/ 7 Layer

DHCP Message

Operation Code 1 Byte	Hardware Type 1 Byte	Hardware Length 1 Byte	Hop Count 1 Byte
Transaction ID 4 Bytes			
Seconds 2 Bytes		Flags 2 Bytes (first bit for broadcast flag)	
Initial Client IP address (CIAddr)			
DHCP Server Issued IP address (YIAddr)			
DHCP Server IP address (SIAddr)			
Gateway IP address (GIAddr)			
Client Hardware Address (CHAddr) 16 Bytes			
Server Name (SName) 64 Bytes			
Boot Filename 128 Bytes			
Options Variable Size			

Initial DHCP Exchange



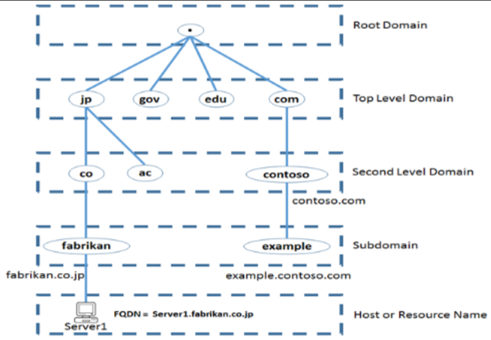
DNS Query message

ID 2 Bytes	Flags 2 Bytes	Number of Questions 2 Bytes
Number of Answer RRs 2 Bytes	Number of Authority RRs 2 Bytes	Number of Additional RRs 2 Bytes
Questions (Variable length)		

additional DNS Reply fields

Answers (Variable length)
Authority (Variable length)
Additional Info (Variable length)

Type	*Mnemonic*	*Description*
1	A	**Address.** A 32-bit IPv4 address. It converts a domain name to an address.
2	NS	**Name server.** It identifies the authoritative servers for a zone.
5	CNAME	**Canonical name.** It defines an alias for the official name of a host.
15	MX	**Mail exchange.** It redirects mail to a mail server.
28	AAAA	**Address.** An IPv6 address.



Transport/ 4 Layer

TCP

Source Port 2 Bytes			Destination Port 2 Bytes
Sequence Number 4 Bytes			
Acknowledgement Number 4 Bytes			
hlen 4 bits	reserved 6 bits	control flags 9 bits	Window size 2 Bytes
Checksum 2 bytes			Urgent Pointer 2 Bytes
Optional Data 0-40 Bytes			

UDP

Source Port 2 Byte	Destination Port 2 Byte
Length 2 Bytes	Checksum 2 Byte

Datagram	UDP Header 8 Bytes	
Segment	TCP Header 20-60 Bytes	Data

Network/ 3 layer

ICMP

Type 1 Byte	Code 1 Byte	Checksum 2 Bytes
Extended header 4 bytes		
Data variable length		

Type:
0x08 Echo Request
0x00 Echo Reply

OSPF

Version 1 Byte	Type 1 Byte	Packet Length 2 Bytes
Router ID 4 Bytes		
Area ID 4 Bytes		
Checksum 2 Bytes		Authentication Type 2 Byes
Authentication 8+ Bytes		

OSPF Header	OSPF Message

Field Values:
Version: 0x02 for OSPFv2
Type:
0x01 Hello
0x02 Database Description (DBD)
0x03 Link State Request (LSR)
0x04 Link State Update (LSU)
0x05 Link State Acknowledge (LSAck)

IPv4 Header

version 4 bits	Header Len 4 bits	Service 8 bits	Total Packet Length 16 bits	
Packet fragmentation ID 16 bits			Flags 3 bits	Fragmentation Offset 13 bits
Time-To-Live 8bits		Protocol 8bits	Header Checksum 16 bits	
Source IP address 4 Bytes				
Destination IP address 4 Bytes				
Option (depreciated)				

Protocol: 01 for ICMP

IP Packet

Header 20-60 Bytes	Data

20-65,535 Bytes

OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport
Network	Network/internet
Data Link	Network Access
Physical	

Data Link / 2 Layer

BPDU

Protocol ID 2 Bytes	Version 1 Byte	Message Type 1 Byte	Flags 1 Byte	Root ID (8 Bytes)	Root Path Cost 4 Bytes	Bridge ID (8 Bytes)	Port ID 2 Bytes	Message Age 2 Bytes	Maximum Age 2 Bytes	Hello time 2 Bytes	Forward Delay 2 Bytes

Blue: Generic Ethernet Frame
Orange: IEEE 802.3 Frame Fields + Generic Ethernet Frame
Yellow: Ethernet Frame with VLAN Tag
Green: 802.2 Frame Fields + Generic Ethernet Frame
Pink: BPDU fields used in Election

ARP Request/Reply

Hardware Type 2 Byes (0x0001 for Ethernet)		Protocol Type 2 Bytes (0x0800 for IP)
Hardware Length 1 Byte (0x04 for MAC)	Protocol Length 1 Byte (0x04 for IPv4)	Operation 2 Bytes (Request 0x0001,Reply 0x0002)
Sender MAC address 6 Bytes (Variable)		
Sender IP address 4 Bytes (Variable)		
Target MAC address 6 Bytes (Variable)		
Target IP address 4 Bytes (variable)		

VLAN Tag

Type ID 16 bits / 2 Bytes	Priority 3 bits	Flag 1 bit	VLAN ID 12 bits

MAC Address

Organizationally Unique ID (OUI) 3 Bytes / 6 Hex	Vendor Assigned Unique ID (NIC/interface) 3 Bytes / 6 Hex

IEEE 802.2 LLC						DSAP 1 Byte	SSAP 1 Byte	CTRL 1 Byte		
IEEE 802.3	SOF 1 Byte				Length 2 Bytes				Data <1500 Bytes	
Ethernet II Frames	Preamble 8 Bytes	Destination Address 6 Bytes	Source Address 6 Bytes	VLAN Tag 4 Bytes	Type 2 Bytes				Data >1536 Bytes	FCS 4 Bytes

Type:0x806 ARP

Subnet Mask Chart

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<tr><td>/24</td><td>/25</td><td>/26</td><td>/27</td><td>/28</td><td>/29</td><td>/30</td></tr>
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